

Andrew Liu

(408) 757-4717 | aliu3203@ucla.edu | [linkedin.com/in/andrew-liu-760187265](https://www.linkedin.com/in/andrew-liu-760187265) | aliu3203.github.io

EDUCATION

University of California, Los Angeles — B.S. Computer Science + Physics, 4.0/4.0 GPA **Expected June 2029**

Relevant Coursework: Data Structures, Graph Theory, Computer Organization, Honors Abstract Algebra, Beams + Accelerator Physics

Awards: USA Computing Olympiad, Gold Division (2026); USA Physics Olympiad, Silver Medal (2025); Putnam Exam, score 13 (2025)

WORK EXPERIENCE

Upscale AI — Software Engineer Intern **June 2026 – Present**

- Productionized a GPU timing engine on an internal cloud platform (**Go** control plane, **OpenAPI/React** frontend, Terraform + Packer GPU AMIs) with fail-closed handling of malformed tool output and content-digest provenance for reproducible builds
- Replaced thousands of brute-force re-simulations with a gradient-driven optimization loop screening ~100 candidates per pass in ~28s; validated correctness with a 19/19 directional sign test and root-caused a regression to a stale baseline
- Cut end-to-end jobs to ~20 minutes by collapsing a **Dockerized** EC2 GPU worker and a remote tool host onto one machine, with a self-validating Tcl/Python extraction pipeline running in the live session

RESEARCH EXPERIENCE

Huang Group, UCLA — Undergraduate Research Assistant **February 2026 – Present**

- Built an unattended **Linux/Bash** daemon supervising 5 concurrent simulations that produced ~20 GB of data over multiple days with zero manual intervention — process-liveness checks, race-free harvesting into a versioned dataset, and auto-restart of finished workers
- Engineered a **C++** pipeline over 500M+ records: two-pass streaming design with pre-allocated buffers for an $O(N^2)$ stage (~ 5×10^8 evaluations); cut wall-clock time ~10× by tuning columnar I/O to deserialize only referenced fields

PROJECTS

IMC Prosperity 4 Trading Bot — Top 250 of 18,000 teams globally **2026**

- Implemented EMA mean-reversion and market-making strategies in **Python**, tuning signals against high-volume historical market data
- Modeled implied-volatility smiles to price options and built an interactive backtest visualizer to speed signal discovery and strategy iteration

Redis Server Clone — Concurrent In-Memory Data Store in Java **2026**

- Built a multithreaded, Redis-compatible server implementing the **RESP wire protocol over TCP** in **Java 21**, one **virtual thread** per client, sharding the keyspace across 64 **ReentrantLock** stripes over a **ConcurrentHashMap** for parallel writes
- Implemented blocking BLPOP with a lock-free compare-and-swap handoff (**AtomicReference**) that delivers each value to exactly one waiter under the push/timeout race and recovers cleanly from mid-block client disconnects
- Backed it with a **Maven** build and 56-test **JUnit 5** suite — including 50-thread × 1,000-op concurrency stress tests asserting no lost or duplicated values — run on every push by **GitHub Actions** CI

Accelerator Literature Classifier — Musumeci Group, UCLA **2026**

- Built a supervised text-classification pipeline in **Python/scikit-learn** (TF-IDF + logistic regression) that mines the Crossref API to label research papers, tuning precision/recall against a hand-corrected ground-truth set with a confusion-matrix harness

Racked — Social Discovery App for Local Thrift Stores **2026**

- Built and shipped an iOS app in **React Native + TypeScript** — 20 screens (geolocated discovery feed/map, social graph, 1:1/group chat, notifications) — on TanStack Query with optimistic cache updates, **281 tests**, and **GitHub Actions** CI
- Found and fixed a side-channel where blocked users' message timing leaked through inbox ordering, replacing a trigger-maintained sort column with a block-filtered lateral join

SKILLS

Languages: C++, Python, Java, C, Go, JavaScript/TypeScript, SQL, Tcl, Assembly

Systems & Infrastructure: Linux/Bash, AWS (EC2, AMIs), Docker, Packer, Terraform, PostgreSQL/PostGIS, REST/OpenAPI, Git, GitHub Actions, Maven

Frameworks & Tools: React, React Native, scikit-learn, PyTorch, pandas, NumPy, JUnit, Jest, ROOT, gdb